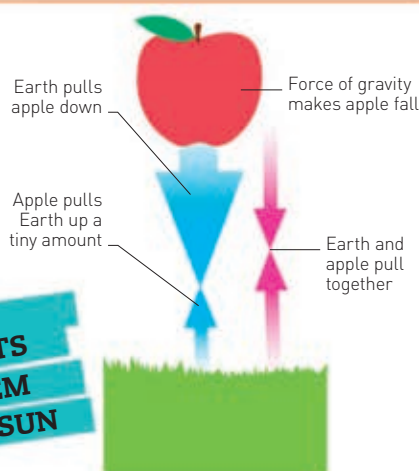


Physics

How do forces, such as gravity and magnetism, affect matter – the stuff all around us? And how does energy make that possible? The answers to these questions are found in physics. Physicists try to unravel the rules of the Universe to explain why the world works as it does.

GRAVITY

Gravity is the force that keeps us held fast on the planet, even while Earth spins at up to 1,670 km/h (1,037 mph). Gravity pulls together all matter, but larger things with more mass have more gravitational force.

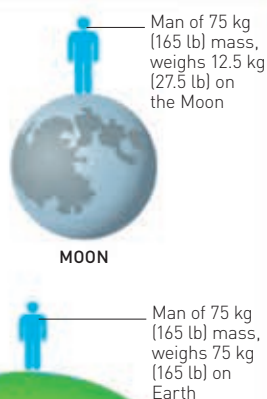


FALLING APPLE

GRAVITY KEEPS EARTH AND THE OTHER PLANETS IN THE SOLAR SYSTEM ORBITING AROUND THE SUN

MASS AND WEIGHT

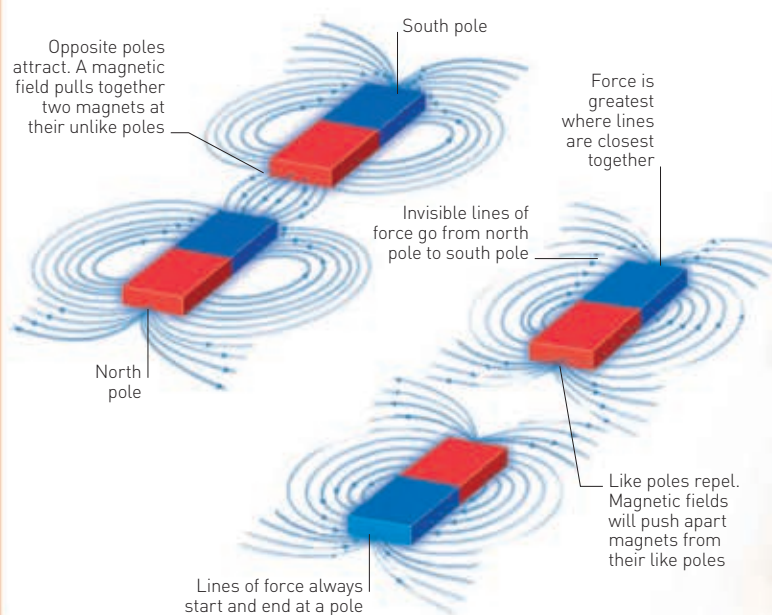
The mass of something is the amount of matter it contains, and mass always stays the same, wherever the object is. But weight changes depending on where an object is, because weight is determined by gravity.



EARTH

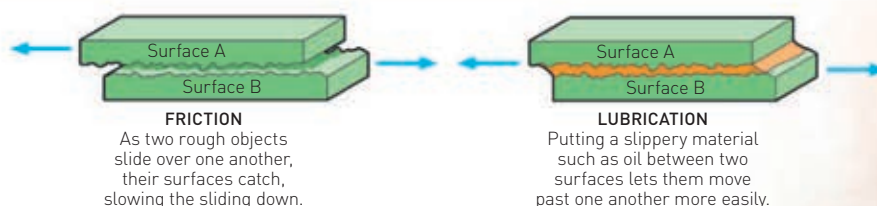
MAGNETISM

Magnetism is a powerful invisible force that is created by electric currents. Magnetic objects have the power to attract other magnetic objects or push them away, depending on how their ends (poles) are lined up.



FRICTION

This force occurs when one object is dragged over the surface of another object. The rougher a surface is, the more friction it produces. Even smooth surfaces have tiny bumps that will produce some friction.



FORCE

A force is something that pushes or pulls objects – whenever something moves, it has been moved by a force. Forces can change the speed of an object, alter its direction, or change its shape.

CHANGING SPEED

The force of the golf club hitting the ball makes the ball move. The ball gains energy and takes off down the golf course.

The harder the ball is hit, the more force is used, and the further it travels



CHANGING DIRECTION

When a force is applied to a moving object like a tennis ball, it can move it in a different direction.

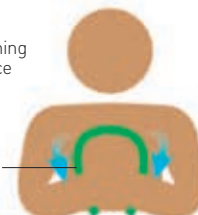
Ball moves in one direction towards the racket
Hitting the ball with the racket applies force, changing the ball's direction



CHANGING SHAPE

A force may cause something to change shape if the force is strong enough and the atoms inside the object cannot resist it.

Bending a bar rearranges the atoms inside it, altering its shape



LAWS OF MOTION

All motion is caused by forces pushing and pulling. The scientist Isaac Newton described three laws of motion. The first says that all things will stay still or move at a steady speed unless a force acts on them. The second says that when a force acts on something it makes it accelerate. The third says that when a force operates on something (action), there is always an opposing and equal force (reaction).

FIRST LAW
Before take-off, the only force acting on a rocket is gravity.

SECOND LAW
The main engines and booster rockets create a huge downward force that accelerates the rocket upwards.

THIRD LAW
The exhaust gas firing down (the action) makes the rocket shoot up (the reaction). The rocket does not push against the air: it moves up because of the force of the exhaust blasting down.

